

Azure AI Search Retrieval Validation Checklist

Corporate Policy RAG

Keyword/BM25, Vector, Hybrid, and Hybrid + Semantic Ranker

Use this checklist to verify that each retrieval mode is configured correctly and that the expected policy documents and chunks are returned. The supplied search output already shows populated chunks, metadata, vector embeddings, and semantic-ranking fields, so this phase focuses on retrieval validation.

1. Pre-Test Validation

- Open Azure Portal -> Azure AI Search service -> Indexes -> select the RAG index -> Search Explorer.
- Switch Search Explorer to JSON view.
- Verify the index contains the fields used by these tests: title, chunk, parent_id, and text_vector.
- Verify the text_vector field is populated for indexed chunks.
- Verify the expected policy documents are present in the index.

2. Test 1 - Validate Keyword / BM25 Retrieval

Run this query:

```
{
  "search": "What is the corporate remote work policy?",
  "select": "title,chunk,parent_id",
  "top": 5
}
```

- **Verify the search field exists.** It should contain the user question: "What is the corporate remote work policy?"
- **Verify the search field contains the exact user question.** This confirms the request will perform lexical/BM25 retrieval against searchable text fields.
- **Verify select includes title, chunk, and parent_id.** These fields let you identify the source document and inspect the retrieved content.
- **Verify top is set to 5.** This limits the final result set to the five highest-ranked keyword matches.
- **Run the query and inspect the returned titles.** remote-work-policy.pdf should rank at or near the top.
- **Inspect the returned chunk.** It should contain substantive remote-work policy content, not only the title or metadata.

PASS when: the expected remote-work document appears near the top and the chunk contains relevant policy content.

3. Test 2 - Validate Vector Retrieval

Run this query:

```
{
  "vectorQueries": [
    {
      "kind": "text",
      "text": "What is the corporate remote work policy?",
      "fields": "text_vector",
      "k": 5
    }
  ],
  "select": "title,chunk,parent_id",
  "top": 5
}
```

- **Verify vectorQueries exists.** This confirms vector retrieval is included in the request.
- **Verify kind is set to text.** This tells Azure AI Search to vectorize the natural-language query at query time, assuming an index vectorizer is configured.
- **Verify vectorQueries.text contains the user question.** This is the text that will be converted into a query embedding.
- **Verify fields is set to text_vector.** This points the vector query at the indexed embedding field.
- **Verify k is set.** For this isolated vector test, k = 5 requests the five nearest vector matches.
- **Verify select includes title, chunk, and parent_id.** This lets you evaluate the source and content returned by vector similarity.
- **Verify top is set to 5.** This returns the five final results for inspection.
- **Run the query and inspect the returned titles and chunks.** remote-work-policy.pdf should rank highly even when wording differs from the source text.

PASS when: the vector query returns semantically relevant remote-work content and the expected policy ranks near the top.

4. Test 3 - Validate Hybrid Retrieval

Run this query:

```
{
  "search": "What is the corporate remote work policy?",
  "vectorQueries": [
    {
      "kind": "text",
      "text": "What is the corporate remote work policy?",
      "fields": "text_vector",
      "k": 50
    }
  ],
  "select": "title,chunk,parent_id",
  "top": 5
}
```

- **Verify the search field exists and contains the user question.** This confirms the request includes keyword/BM25 retrieval.
- **Verify vectorQueries exists.** This confirms vector retrieval is included in the same request.
- **Verify vectorQueries.text contains the same user question.** This confirms the user question will also be vectorized for semantic similarity retrieval.
- **Verify kind is set to text.** This enables query-time text vectorization when the configured vectorizer supports it.
- **Verify fields is set to text_vector.** This identifies the indexed vector field used for similarity matching.
- **Verify k is set to 50.** This provides a broader vector candidate set before the hybrid ranking is combined.
- **Verify select includes title, chunk, and parent_id.** These fields allow you to validate which source and chunk were selected.
- **Verify top is set to 5.** This returns the five highest-ranked final hybrid results.
- **Run the query and compare the ranking with the keyword-only and vector-only tests.** The expected policy should remain at or improve toward the top.

PASS when: the request clearly contains both BM25 and vector retrieval, and the combined results rank relevant policy chunks at the top.

5. Test 4 - Validate Hybrid + Semantic Ranker

Run this query:

```
{
  "search": "What is the corporate remote work policy?",
  "vectorQueries": [
    {
      "kind": "text",
      "text": "What is the corporate remote work policy?",
      "fields": "text_vector",
      "k": 50
    }
  ],
  "queryType": "semantic",
  "semanticConfiguration": "YOUR-SEMANTIC-CONFIGURATION-NAME",
  "captions": "extractive",
  "answers": "extractive",
  "select": "title,chunk,parent_id",
  "top": 5
}
```

- **Verify search and vectorQueries are both present.** This confirms the request remains hybrid.
- **Verify vectorQueries.text contains the user question.** This confirms the same question is used for vectorization.
- **Verify fields is set to text_vector and k is set.** This confirms the vector search target and candidate count.
- **Verify queryType is set to semantic.** This enables semantic ranking for the retrieved candidates.

- **Verify semanticConfiguration is set to the actual configuration name for the index.** Do not leave the placeholder value in the final test.
- **Verify captions is set to extractive.** This requests a concise passage that helps you judge why a result is relevant.
- **Verify answers is set to extractive.** This requests an extractive answer when the query and semantic configuration support one.
- **Verify top is set to 5.** This limits the final result set to five results.
- **Inspect @search.rerankerScore, captions, answers, titles, and chunks.** Use these together to judge relevance; do not evaluate success from @search.score alone.

PASS when: the correct policy is highly ranked and semantic reranking/captions improve the usefulness of the returned context.

6. Validate the Four Representative Policy Questions

Remote work

- **Run:** What is the corporate remote work policy?
- **Expected primary source:** remote-work-policy.pdf
- **Verify:** Verify the expected source ranks near the top and the chunk contains eligibility/security/work-environment guidance.
- **Record:** the source rank (#1, #2, #3, or Not found) for Keyword, Vector, Hybrid, and Hybrid + Semantic.

Password changes

- **Run:** How frequently do employees need to change passwords?
- **Expected primary source:** password-reset-guide.txt
- **Verify:** Verify the source is retrieved, but do not treat a missing fixed rotation interval as a failure. The supplied guide discusses expiration, prompted changes, and immediate changes after suspected exposure.
- **Record:** the source rank (#1, #2, #3, or Not found) for Keyword, Vector, Hybrid, and Hybrid + Semantic.

Incident escalation

- **Run:** What is the incident escalation procedure?
- **Expected primary source:** incident-management-process
- **Verify:** Verify the incident-management source ranks highly and the returned chunk contains actual escalation procedure details.
- **Record:** the source rank (#1, #2, #3, or Not found) for Keyword, Vector, Hybrid, and Hybrid + Semantic.

Cloud costs

- **Run:** What policies apply to cloud costs?
- **Expected primary source:** cloud-cost-policy.pdf
- **Verify:** Verify the cloud-cost policy ranks at or near the top and the chunk contains cost governance requirements such as ownership, tags, budgets, alerts, optimization, or enforcement.
- **Record:** the source rank (#1, #2, #3, or Not found) for Keyword, Vector, Hybrid, and Hybrid + Semantic.

7. Validate Paraphrased Questions

Run paraphrased versions to confirm semantic retrieval is working beyond exact title/keyword matches.

- **Run:** Can I work from home and what security rules do I have to follow?
 - **Verify expected source:** remote-work-policy.pdf
- **Run:** Who should I contact when a serious production issue gets worse?
 - **Verify expected source:** incident-management-process
- **Run:** What are the rules for controlling Azure and cloud spending?
 - **Verify expected source:** cloud-cost-policy.pdf

PASS when: hybrid/vector retrieval still finds the correct policy even when the query uses different wording.

8. Record Retrieval Rankings

Query	Keyword	Vector	Hybrid	Hybrid + Semantic
Remote work	Rank:	Rank:	Rank:	Rank:
Password frequency	Rank:	Rank:	Rank:	Rank:
Incident escalation	Rank:	Rank:	Rank:	Rank:
Cloud costs	Rank:	Rank:	Rank:	Rank:
Paraphrase tests	Rank:	Rank:	Rank:	Rank:

Record values such as #1, #2, #3, or Not found.

9. Final Validation Checklist

- Keyword/BM25 query returns relevant policy documents.
- Vector query uses vectorQueries and the text_vector field.
- Hybrid query contains both search and vectorQueries.
- Hybrid vectorQueries.text contains the user's natural-language question.
- Hybrid query has an appropriate k value and top is explicitly set.
- Hybrid + Semantic query uses the correct semantic configuration.
- Expected source documents rank near the top for representative questions.
- Returned chunks contain enough relevant content for the LLM to answer.
- Paraphrased questions still retrieve the correct policy.
- The system does not invent policy details that are absent from the source content.

10. Completion Criteria

Mark the Vector and Hybrid Retrieval testing phase complete when the expected source consistently ranks near the top for exact and paraphrased questions, the returned chunks contain useful policy context, and the Hybrid + Semantic configuration produces stable, relevant results suitable for the RAG application.